

INVERCARGILL, New Zealand

WMO: 938450

Lat: 46.4108S

Lon: 168.3178E

Elev: 3

StdP: 101.29

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.2	-0.9	-3.1	2.9	-1.6	-2.0	3.2	-0.1	14.5	7.2	13.1	7.1	1.6	30	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.0	23.6	17.0	21.9	16.4	20.3	15.5	18.1	22.2	17.0	20.9	16.1	19.4	5.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.3	11.6	20.0	15.3	10.9	18.8	14.5	10.3	17.9	51.3	22.2	47.9	20.9	45.2	19.6	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.8	13.2	11.6	DB	-4.0	27.5	1.1	1.7	-4.8	28.7	-5.4	29.6	-6.0	30.6	-6.8	31.8	(4)
(5)			WB	-4.2	20.0	1.1	1.0	-5.0	20.7	-5.7	21.3	-6.3	21.9	-7.1	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	14.0	14.0	12.7	10.6	8.5	6.3	5.8	6.8	8.5	9.7	11.1	13.1	(6)	
	DBStd	3.78	2.62	2.65	2.61	2.51	2.52	2.38	2.39	2.23	2.39	2.41	2.65	2.66	(7)	
	HDD10.0	543	1	1	4	22	61	113	130	102	57	35	16	3	(8)	
	HDD18.3	3022	137	124	174	232	306	361	387	358	296	267	217	163	(9)	
	CDD10.0	571	124	113	89	40	13	2	1	2	12	26	49	99	(10)	
	CDD18.3	7	3	3	1	0	0	0	0	0	0	0	0	1	(11)	
	CDH23.3	66	26	20	5	0	0	0	0	0	0	0	2	13	(12)	
	CDH26.7	6	3	2	0	0	0	0	0	0	0	0	0	1	(13)	
(14) Wind	WSAvg	4.7	5.5	4.8	4.6	4.4	4.2	4.1	3.8	3.9	4.8	5.6	5.6	5.2	(14)	
(15) Precipitation	PrecAvg	1088	106	79	94	96	112	101	81	71	81	90	88	96	(15)	
	PrecMax	1444	215	165	211	192	192	201	173	162	173	163	162	230	(16)	
	PrecMin	812	37	13	28	16	45	29	29	16	10	31	22	33	(17)	
	PrecStd	138	49	39	44	47	36	38	33	31	39	33	39	42	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.5	25.8	24.0	20.4	17.6	14.2	13.7	15.2	18.4	20.6	22.5	25.2	(19)	
		MCWB	18.8	18.5	17.0	15.1	13.3	10.7	10.0	10.7	12.4	14.4	16.1	17.5	(20)	
	2%	DB	23.1	23.2	21.2	18.2	15.4	12.2	12.0	13.2	16.1	18.1	20.0	22.5	(21)	
		MCWB	17.0	17.3	16.4	13.7	12.1	9.7	9.0	9.7	11.3	13.1	15.0	16.3	(22)	
	5%	DB	21.0	21.1	19.5	16.6	13.9	11.1	10.7	11.9	14.4	16.1	18.1	20.2	(23)	
		MCWB	15.9	16.3	15.4	12.9	11.4	9.3	8.5	9.0	10.5	12.0	13.7	15.2	(24)	
	10%	DB	19.2	19.2	17.7	15.1	12.7	10.3	9.6	10.8	12.9	14.4	16.0	18.3	(25)	
		MCWB	15.2	15.5	14.5	12.2	10.6	8.7	7.8	8.5	9.8	11.1	12.4	14.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.3	19.7	18.6	15.8	14.0	11.3	10.6	11.1	13.1	15.1	17.1	18.4	(27)	
		MCDB	24.5	24.6	22.2	19.1	16.3	13.3	12.9	14.1	17.1	19.3	20.9	23.6	(28)	
	2%	WB	17.9	18.0	17.0	14.3	12.6	10.2	9.5	10.1	11.8	13.4	15.3	16.9	(29)	
		MCDB	21.8	22.1	20.1	17.0	14.7	11.6	11.2	12.5	15.1	17.1	19.2	21.2	(30)	
	5%	WB	16.6	16.8	15.8	13.3	11.6	9.6	8.7	9.4	10.8	12.4	14.1	15.8	(31)	
		MCDB	20.3	20.4	18.8	15.7	13.5	10.9	10.2	11.4	13.7	15.6	17.2	19.2	(32)	
	10%	WB	15.5	15.7	14.7	12.5	10.8	8.8	8.0	8.7	10.0	11.4	12.9	14.7	(33)	
		MCDB	18.6	18.6	17.4	14.7	12.4	10.2	9.4	10.5	12.5	14.1	15.6	17.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	8.1	8.1	7.7	6.9	6.5	6.9	7.5	7.6	7.5	7.5	8.0	(35)	
		MCDBR	12.2	11.9	11.6	10.5	8.9	7.6	7.8	9.0	10.4	10.8	11.2	11.9	(36)	
	5% WB	MCWBR	7.2	7.1	7.3	6.9	6.5	6.0	5.9	6.4	6.7	7.0	7.0	6.8	(37)	
		MCWBR	10.8	10.7	10.1	8.8	8.1	6.9	6.9	8.0	9.1	9.9	10.2	10.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.343	0.330	0.317	0.309	0.300	0.300	0.291	0.301	0.325	0.327	0.330	0.330	(40)	
		taud	2.501	2.555	2.593	2.597	2.584	2.532	2.579	2.580	2.501	2.511	2.515	2.545	(41)	
	Ebn at Noon		975	958	922	850	772	719	773	846	896	952	983	996	(42)	
		Edh at Noon	109	97	84	70	57	54	56	69	90	101	107	106	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.76	4.99	3.63	2.31	1.35	1.03	1.26	2.01	3.18	4.50	5.52	6.12	(44)	
	RadStd		0.43	0.33	0.18	0.14	0.09	0.08	0.09	0.12	0.18	0.29	0.36	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	+0.65	N/A	N/A	N/A	-53	-132	+76	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	+0.70	N/A	N/A	N/A	-47	-112	N/A	N/A	N/A

Nomenclature: See separate page